Automated Email Phishing Detection Using Machine Learning and MLOps Pipeline

Amar H. M.
Department of AI & ML
Dayananda Sagar University
Bengaluru India
eng23am1001@dsu.edu.in

Sanjay S.
Department of AI & ML
Dayananda Sagar University
Bengaluru India
eng23am1006@dsu.edu.in

Deepak Mahto.
Department of AI & ML
Dayananda Sagar University
Bengaluru India
eng22am0087@dsu.edu.in

Shashanka N.
Department of AI & ML
Dayananda Sagar University
Bengaluru India
eng23am1006@dsu.edu.in

Index Terms

Phishing Detection, Machine Learning, Logistic Regression, TF–IDF, Email Security, IMAP, FastAPI, MLOps, Prefect.

Abstract

Phishing attacks are a growing cybersecurity threat, exploiting user trust to steal sensitive information. Manual detection methods are no longer sufficient due to the increasing volume and sophistication of malicious emails. This work presents an automated Email Phishing Detection System that performs real- time identification of suspicious emails using classical machine learning techniques. Emails are fetched through an IMAP pipeline, preprocessed to remove noise, and transformed using TF–IDF features. A Logistic Regression classifier trained on 1,000 labeled emails predicts phishing probability and assigns labels using an optimized threshold. The system includes logging for performance monitoring, a FastAPI interface for seamless integration, and Prefect-based automation for scheduled fetching, prediction, and retraining. The solution is scalable, adaptive, and reduces human intervention while strengthening email security.

I. Introduction

Email remains a fundamental mode of communication across personal, organizational, and governmental domains. Its widespread use, however, has made it a prime target for cyberattacks, particularly phishing. Phishing emails impersonate trusted entities to deceive recipients into revealing sensitive information, clicking malicious links, or downloading harmful attachments. Such attacks can lead to financial loss, identity theft, data breaches, and operational disruption.

With increasing sophistication, phishing campaigns now employ advanced social engineering, personalized content, and spoofed identities, allowing them to evade traditional defense mechanisms. Rule-based filters, blacklists, and manual inspection are no longer sufficient, especially given the high volume of incoming emails and the ability of attackers to circumvent static detection methods through minor content modifications.

Machine learning offers a promising direction by identifying hidden patterns in email content and metadata that distinguish phishing from legitimate messages. ML-based systems can detect subtle cues, learn from historical data, and adapt to evolving attack techniques. However, implementing such a system requires coordinated components for email retrieval, preprocessing, feature extraction, model training, prediction, automation, and monitoring.

This project presents a complete Email Phishing Detection System integrating these requirements. The system retrieves emails via IMAP, extracts key metadata, preprocesses textual content, and converts it into numerical representations using TF–IDF features. A Logistic Regression model trained on 1,000 labeled emails predicts phishing probabilities and assigns labels using an optimized threshold. Predictions are logged for monitoring and analysis.

To support continuous improvement, an automated MLOps pipeline built with Prefect schedules email fetching, prediction, and retraining. A FastAPI-based interface provides real-time predictions for easy integration with existing applications. Together, these components deliver a scalable, adaptive, and reliable solution that enhances phishing detection accuracy while reducing manual effort.

II. Literature Review

Phishing detection has been widely studied using both traditional and machine-learning-based techniques. Early approaches relied on rule-based filters, blacklists, and header analysis to identify malicious emails. While effective for known threats, these methods fail against new or obfuscated attacks. Content-based detection using keyword frequency, heuristic rules, and URL pattern analysis also showed limitations due to their static nature.

Machine learning introduced a more adaptive solution. Algorithms such as Naïve Bayes, Support Vector Machines, Random Forests, and Logistic Regression have been used to classify emails based on textual and structural features. TF–IDF and bag-of-words representations proved effective in capturing discriminative patterns in email content. Studies further emphasize the role of preprocessing—removal of URLs, email addresses, punctuation, and noise— in improving model performance.

Deep learning models such as CNNs, RNNs, and transformers have shown strong performance in analyzing complex phishing emails but require larger datasets and more computational power. Classical ML models remain suitable for lightweight, interpretable, and real-time applications. Literature also highlights the need for continuous retraining and automated pipelines to address evolving phishing techniques.

III. Problem Statement

Phishing emails have become increasingly sophisticated, making traditional rule-based filters and manual inspection ineffective. Large email volumes, evolving attack patterns, and subtle social-engineering techniques make accurate identification difficult. There is a need for an automated, adaptive, real-time system that can reliably detect phishing emails using machine learning while reducing human effort and improving security.

IV. Objectives

- Develop an automated system that fetches and analyzes emails in real time.
- Preprocess email text and extract meaningful features using TF–IDF.
- Train a machine learning model to accurately classify phishing and legitimate emails.
- Build a FastAPI interface for real-time phishing prediction.
- Implement an automated Prefect pipeline for periodic email fetching, prediction, logging, and retraining.
- Reduce manual inspection while improving detection accuracy.

V. Dataset Description

The dataset consists of 1,000 labeled emails classified as phishing or legitimate. Each email contains metadata such as sender address, subject, and message body. The dataset combines publicly available phishing corpora and manually collected legitimate emails, ensuring balanced representation. Preprocessing includes removal of URLs, email addresses, numbers, punctuation, and excess whitespace. The cleaned dataset forms the basis for TF–IDF feature extraction and machine learning classification.

VI. Proposed Methodology

The methodology follows a structured pipeline:

1. **Email Retrieval**

   - Fetch emails via IMAP and extract sender, subject, and body.

2. **Preprocessing**

   - Clean text by removing URLs, email addresses, numbers, punctuation, and extra whitespace.

3. **Feature Extraction**

   - Apply TF–IDF vectorization using unigrams and bigrams.

4. **Model Training and Prediction**

   - Train Logistic Regression on 1,000 labeled emails.
   - Predict phishing probability and classify using an optimized threshold.

5. **System Automation and Deployment**

   - Provide real-time predictions through FastAPI.
   - Use Prefect for automated periodic fetching, prediction, logging, and retraining.

VII. System Architecture

A. Overview Components

1. Email Ingestion Layer
2. Text Preprocessing Layer
3. Feature Extraction Layer
4. Machine Learning Classification Layer
5. Logging & Monitoring Layer
6. API & Real-Time Prediction Layer
7. MLOps / Continuous Learning Layer

B. Component Summary

- **Email Ingestion:** Connects to mailbox, retrieves emails, decodes MIME.
- **Preprocessing:** Normalizes text and removes noise.
- **Feature Extraction:** TF–IDF vectorization with unigrams/bigrams.
- **Classification:** Logistic Regression with probability-based labeling.
- **Logging:** Stores predictions and scores.
- **API Layer:** FastAPI endpoint for immediate predictions.

- **MLOps Layer:** Prefect handles scheduled workflow and retraining.

C. Data Flow (Summary)

VIII. Conclusion

This work presents a modular and automated Email Phishing Detection System integrating IMAP-based ingestion, structured preprocessing, TF–IDF feature extraction, and Logistic Regression classification. The system provides accurate detection of malicious emails, reduces manual effort, and adapts to evolving threats through a Prefect-based MLOps pipeline. With its real-time FastAPI interface and scalable design, the solution offers a practical and effective approach to strengthening email security.

IX. Future Work

- Implement deep learning models (LSTM, BERT, transformers) for improved accuracy.
- Expand the dataset using larger and diverse real-world email samples.
- Add URL reputation checks and attachment scanning for deeper threat analysis.
- Integrate authentication protocols such as SPF, DKIM, and DMARC.
- Deploy as a browser extension or integrate with enterprise email servers.
- Add advanced MLOps features like drift detection and automated model versionin

X. ACKNOWLEDGMENT

The authors express their sincere gratitude to **Prof. Shivnandan Rai** for his valuable guidance, encouragement, and insightful feedback throughout the development of this work. The authors also thank the Department of Artificial Intelligence & Machine Learning, Dayananda Sagar University, for providing the necessary resources and support. Their continued motivation and technical assistance played a vital role in the successful completion of this project

XI. REFERENCES

[1] G. Salton and M. J. McGill, *Introduction to Modern Information Retrieval*, McGraw-Hill, 1983.
(Used for TF-IDF foundations)

[2] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed., Wiley, 2013.
(Used for Logistic Regression theory)

[3] S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*, O'Reilly Media, 2009.
(NLTK and preprocessing reference)

[4] A. Jain, B. Gupta, and A. Kumar, "Phishing Email Detection Using Machine Learning Approaches," *International Journal of Computer Applications*, vol. 174, no. 19, pp. 1–9, 2021.
(General ML phishing research)

[5] Symantec Corporation, "Email Threat Report," 2020.
(Industry phishing statistics)

[6] Python Software Foundation, "email — An email and MIME handling package," Python Documentation.
(Used for IMAP email processing)

[7] Prefect Technologies Inc., "Prefect: Workflow Orchestration for Data Applications," 2024.
(Used for MLOps pipeline automation)

[8] FastAPI Documentation, "FastAPI: High-Performance ASGI Framework," 2024.
(Used for API deployment)

[9] Scikit-learn Developers, "Scikit-learn: Machine Learning in Python," 2024.
(Used for TF–IDF and Logistic Regression implementation